CAMBRIDGE INSTITUTE OF TECHNOLOGY
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इलेक्ट्रॉनिकी एवं सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF ELECTRONICS AND INFORMATION TECHNOLOGY
सत्यमेव जयते

Chips to Startup Programme

सी-डैक CDAC

ChipIN

Cambrian Research Park

Day 5 – Session

gm/ID Methodology for Analog IC Design: SCL 180 nm NMOS Characterization

ANGSTROMERS
Navigating Silicon Waves

- **gm (Transconductance)**

The fundamental expression for transconductance is:

$gm = \partial ID / \partial VGS$

For a MOSFET operating in saturation,

- **Drain current**

$ID = (1/2)\ \mu nCox(W/L)(VGS - VTH)^2$

Therefore, the **transconductance** can be expressed as:

$gm = \mu nCox(W/L)(VGS - VTH)$

Alternatively, in terms of drain current:

$gm = \sqrt{[2\mu nCox(W/L)ID]}$

where:μn = Electron mobility and Cox = Oxide capacitance per unit area, W=Channel width, L = Channel length,

VGS = Gate-source voltage,VTH = Threshold voltage ,ID = Drain current.

- Thus, gm indicates the small-signal current-driving capability of the MOSFET and is an important parameter for determining the gain and performance of analog circuits.

- The gm/ID methodology is a transistor-sizing and biasing technique widely used in analog IC design.
- It uses the ratio gm/ID as a measure of transconductance efficiency, indicating how effectively a MOSFET converts bias current into transconductance. By selecting an appropriate gm/ID, the designer can determine the required drain current:
- Thus, the gm/ID methodology provides a systematic approach to transistor biasing and sizing, while allowing the designer to control important trade-offs among power, gain, speed, and area.

Factors Captured by *gm/ID* Analysis

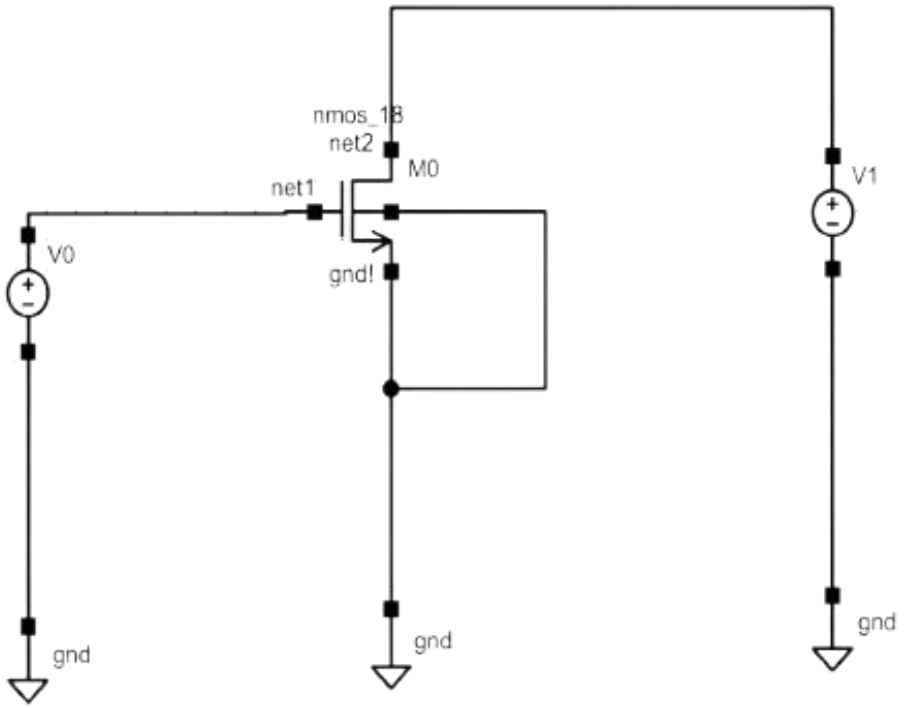
gm/ID analysis captures multiple device and performance parameters, allowing a complete understanding of the trade-offs in transistor operation.

Key Relationship

$$g_m = I_D\left(\frac{g_m}{I_D}\right)$$

Parameter	What it tells you
g_m/I_D	**Transconductance efficiency** — how efficiently bias current is converted into transconductance.
I_D/W	Current density → helps determine the required MOS width (W).
V_{GS}	Required gate-source bias voltage.
$V_{OV} = V_{GS} - V_{TH}$	Overdrive voltage.
g_m/g_{ds}	Intrinsic voltage gain capability.
g_m	Transconductance available for the selected bias point.
f_T	Speed / high-frequency performance.
g_{ds}	Output conductance and channel-length modulation effects.
$r_o = 1/g_{ds}$	Output resistance.
Inversion level	Indicates weak, moderate or strong inversion operation.
Power efficiency	Relates the required current to the desired g_m.
Device sizing	Helps select W/L for a required operating point.

NMOS Test Bench & Device Setup



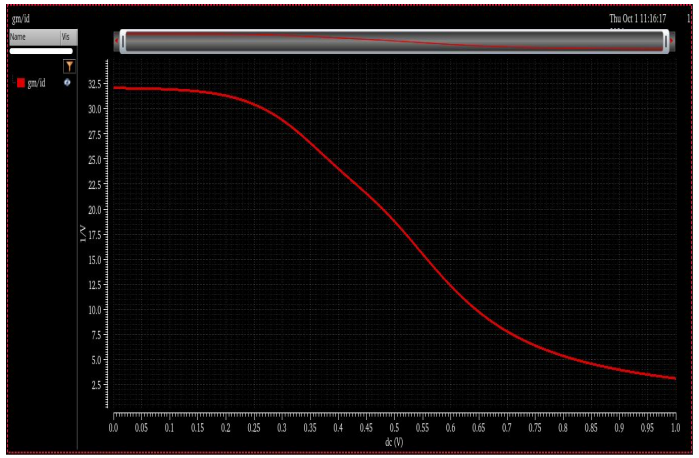
Transistor: Single NMOS (M0)
W = 1µm
L = 0.18µm
Vds = 0.9V
Vgs = 0V to 1V (DC Sweep)
Source = Ground

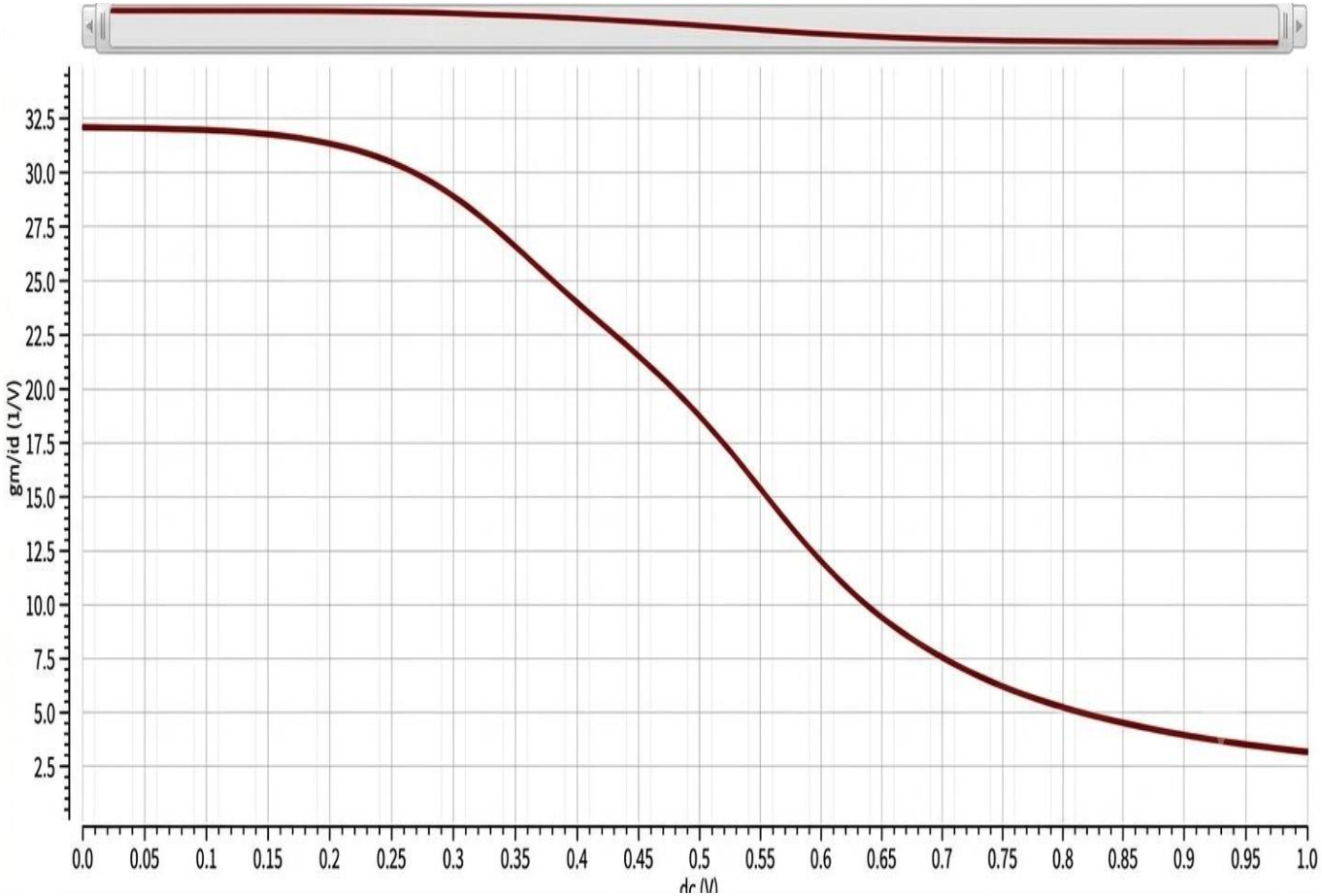
```
save M0:all
```

Purpose: After creating the NMOS setup, the simulator needs explicit instructions to save the internal operating point parameters. To do this, you create a new file called save_op.scs in a text editor, write this single line inside, and save it. This ensures all parameters for transistor M0 are recorded during simulation (which is then loaded by the definition File command in the main script).

Inversion Region	g_m/I_D Range (V^{-1})	VOV = VGS − VTH	Drain Current	Transistor Size	Speed	Power Efficiency	Typical Applications
Weak Inversion (Subthreshold)	20–30+	Very small < ~100 mV	Very low	Very large	Low	Very high	Ultra-low-power circuits, sensors, low-frequency amplifiers
Moderate Inversion	10–20	~100–200 mV	Low–medium	Medium–large	Medium	High	OTA, op-amp, ADC, low-power analog design
Strong Inversion	5–10	~200–400 mV	Medium–high	Small	High	Lower	High-speed amplifiers, buffers, drivers
Very Strong Inversion	<5	Large	High	Small	Very high	Low	High-speed/high-current circuits

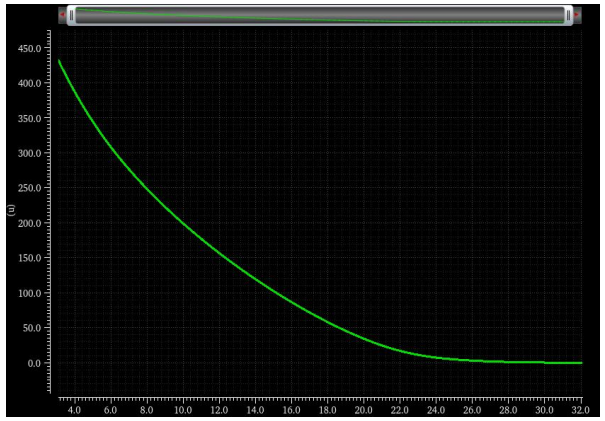
gm/Id vs. Vgs - Transconductance Efficiency Curve

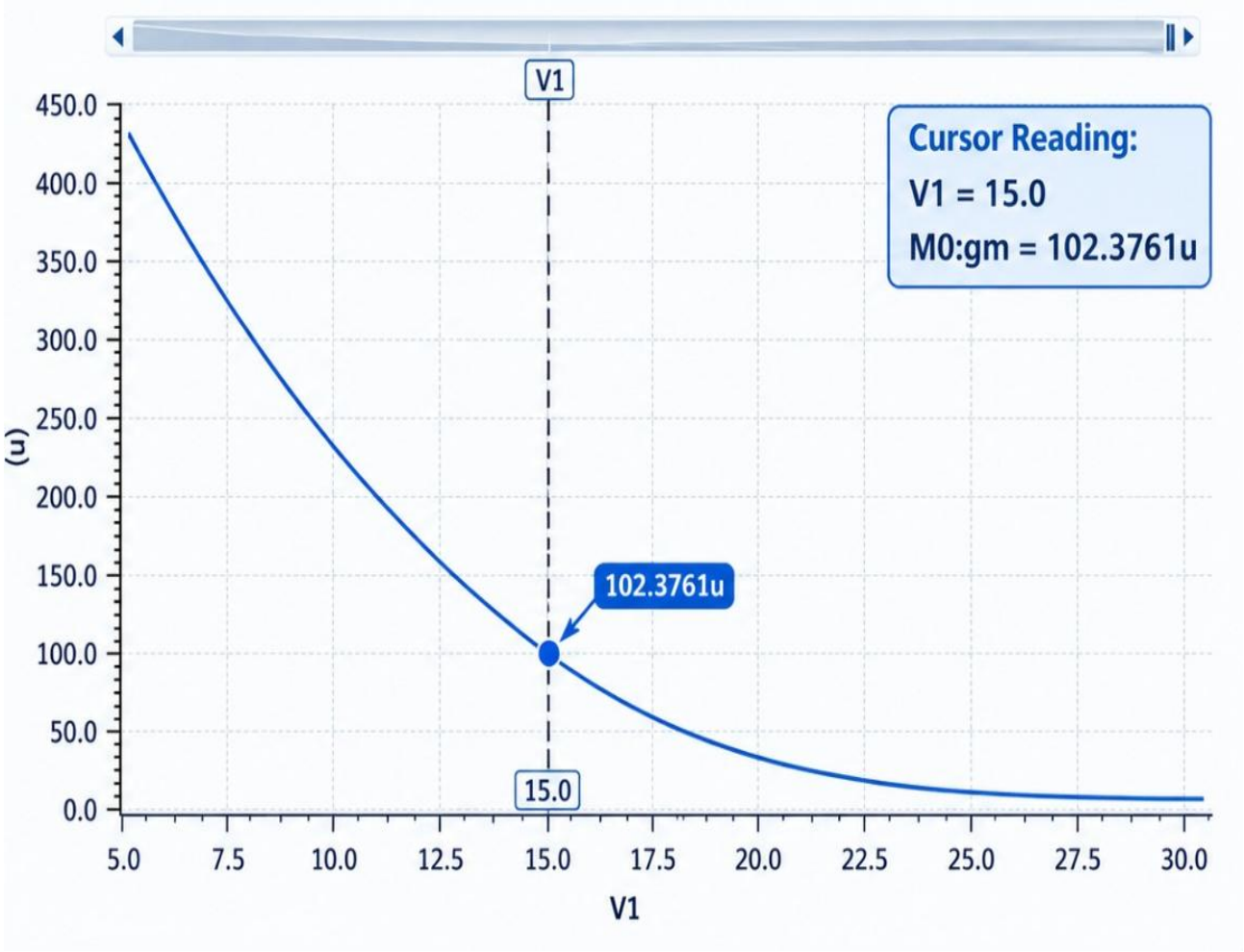




- Y-Axis (gm/Id): Transconductance efficiency—measures how effectively DC bias current converts into gain (gm).
- X-Axis (dc / Vgs): Input sweep voltage controlling the MOSFET's operating region.

gm vs. gm/Id

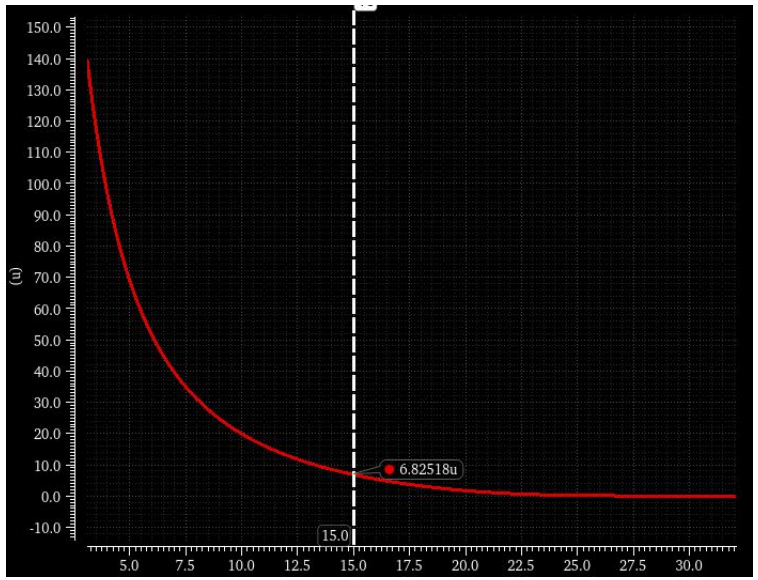




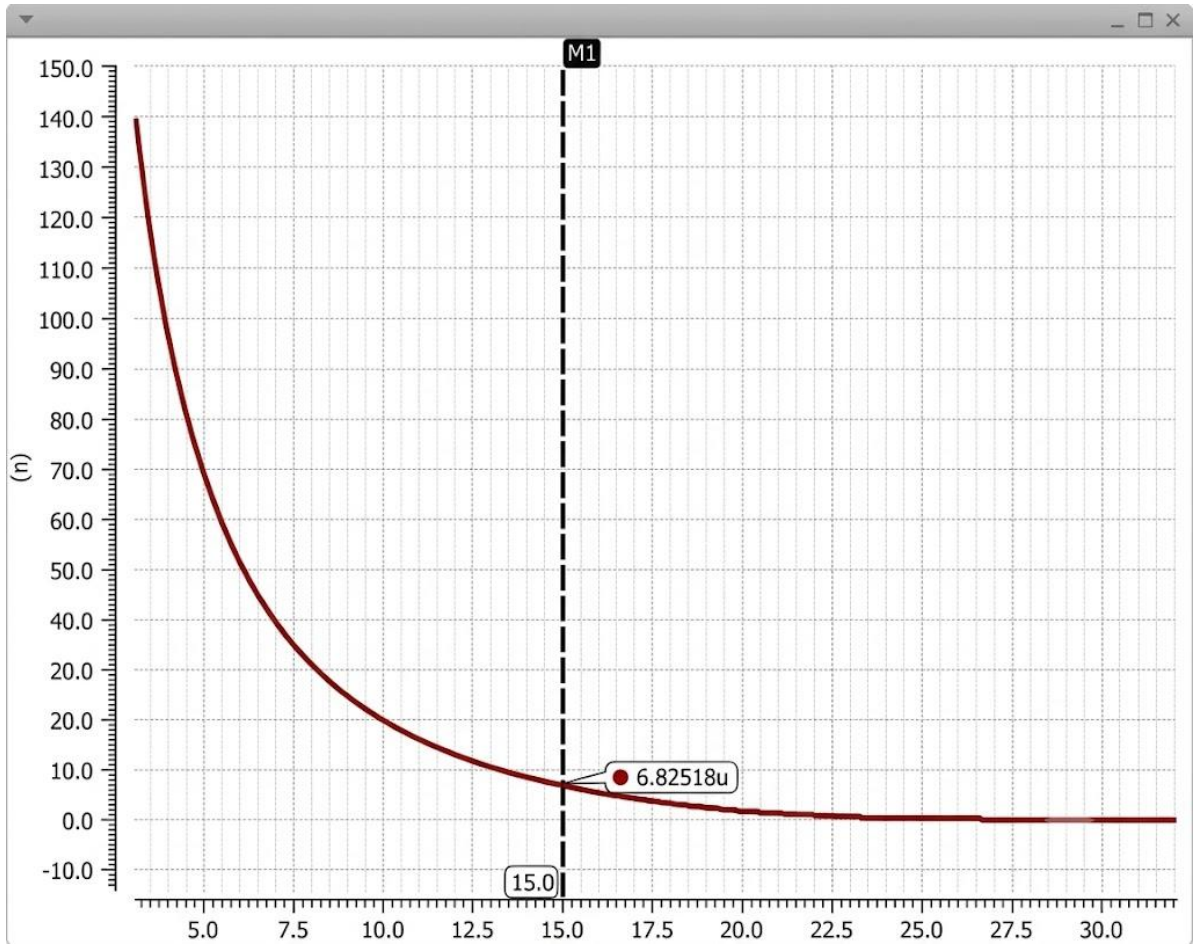
The plot shows transconductance (gm) versus transconductance efficiency (gm/ID) for a fixed transistor size.

As gm/ID increases toward the weak-inversion region, the absolute gm decreases. This illustrates the trade-off between power efficiency and speed.

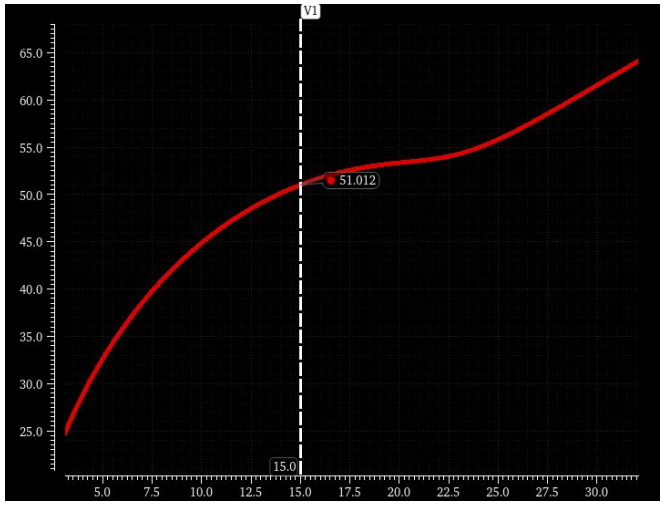
Id vs. gm/Id



At gm/ID = 15 V^{-1}, the transistor provides an Id of approximately 6.825µA. This operating point corresponds to moderate inversion and provides a practical trade-off between power consumption, gain, and speed.

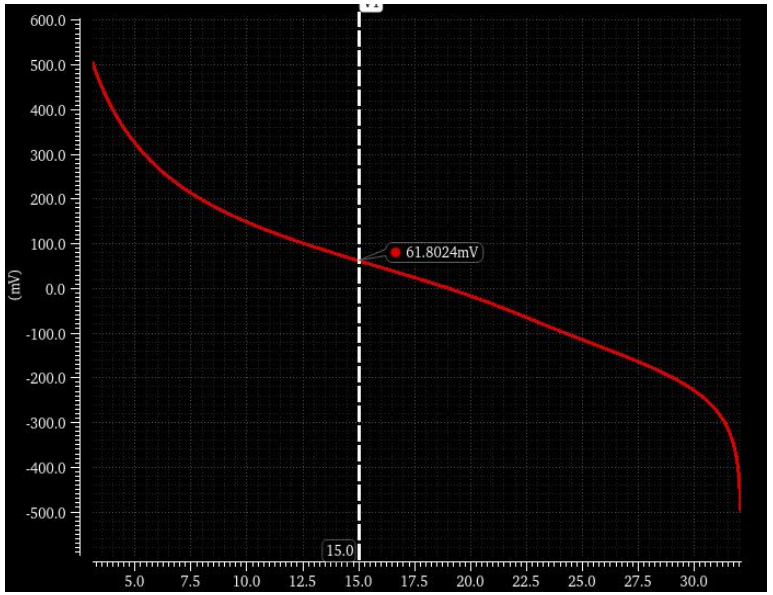


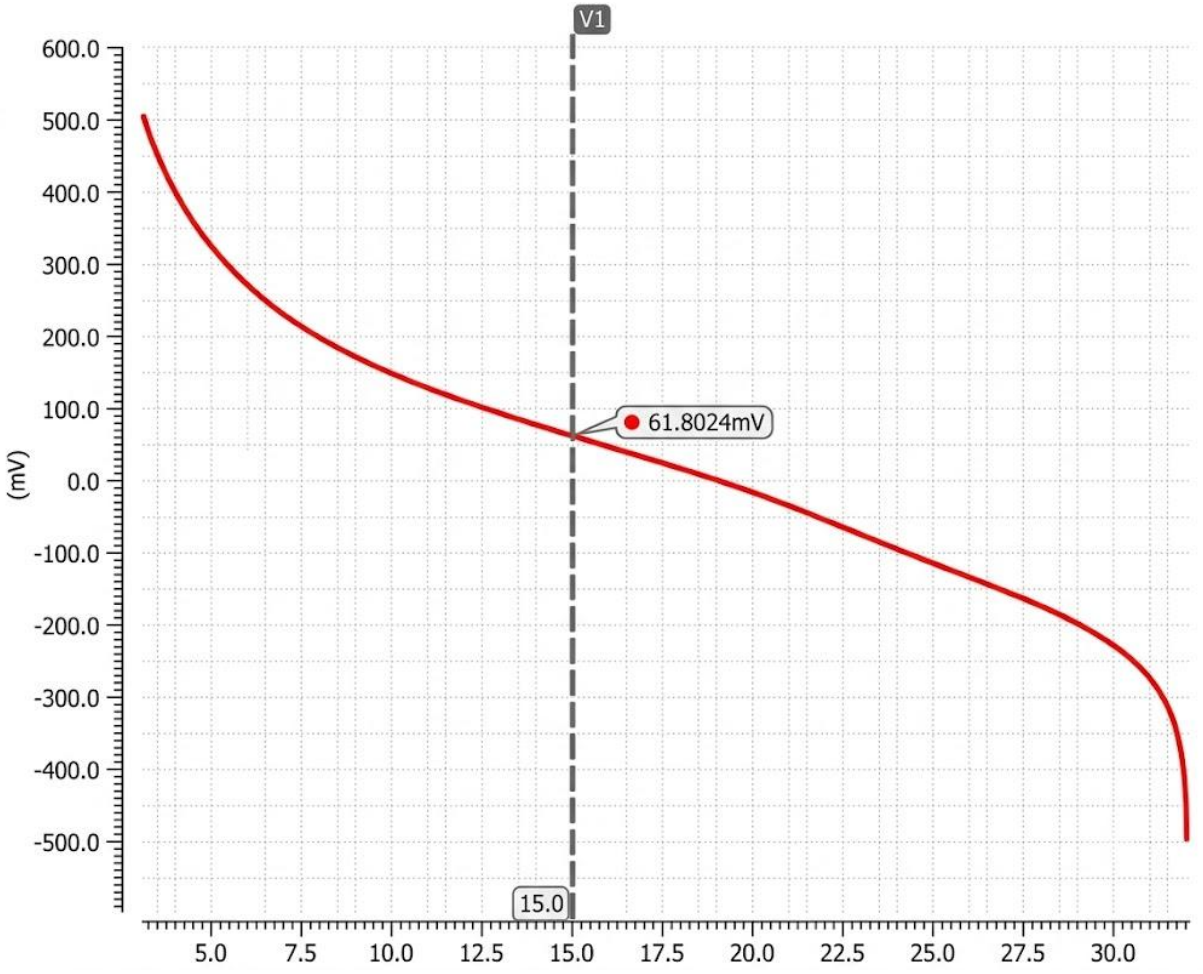
gm/gds (Intrinsic Gain)vs. gm/Id



The transconductance (gm) is defined as:
gm = ∂ID / ∂VGS. The output conductance (gds) is defined as:
gds = ∂ID / ∂VDS.
Therefore, the intrinsic gain is given by:
gm/gds
Since the small-signal output resistance is:
ro = 1/gds

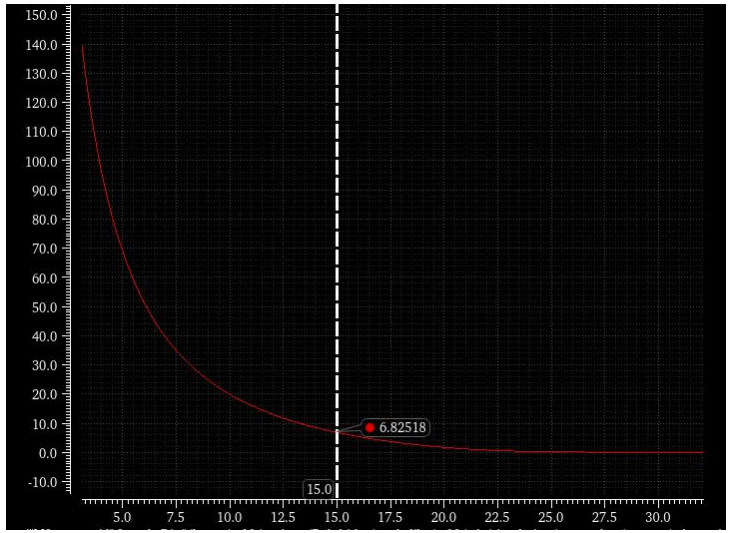
Vov vs. gm/Id





The overdrive voltage is defined as:
VOV = VGS − VTH
where:
Vov =Overdrive voltage
VGS = Gate-source Voltage
VTH = Threshold voltage

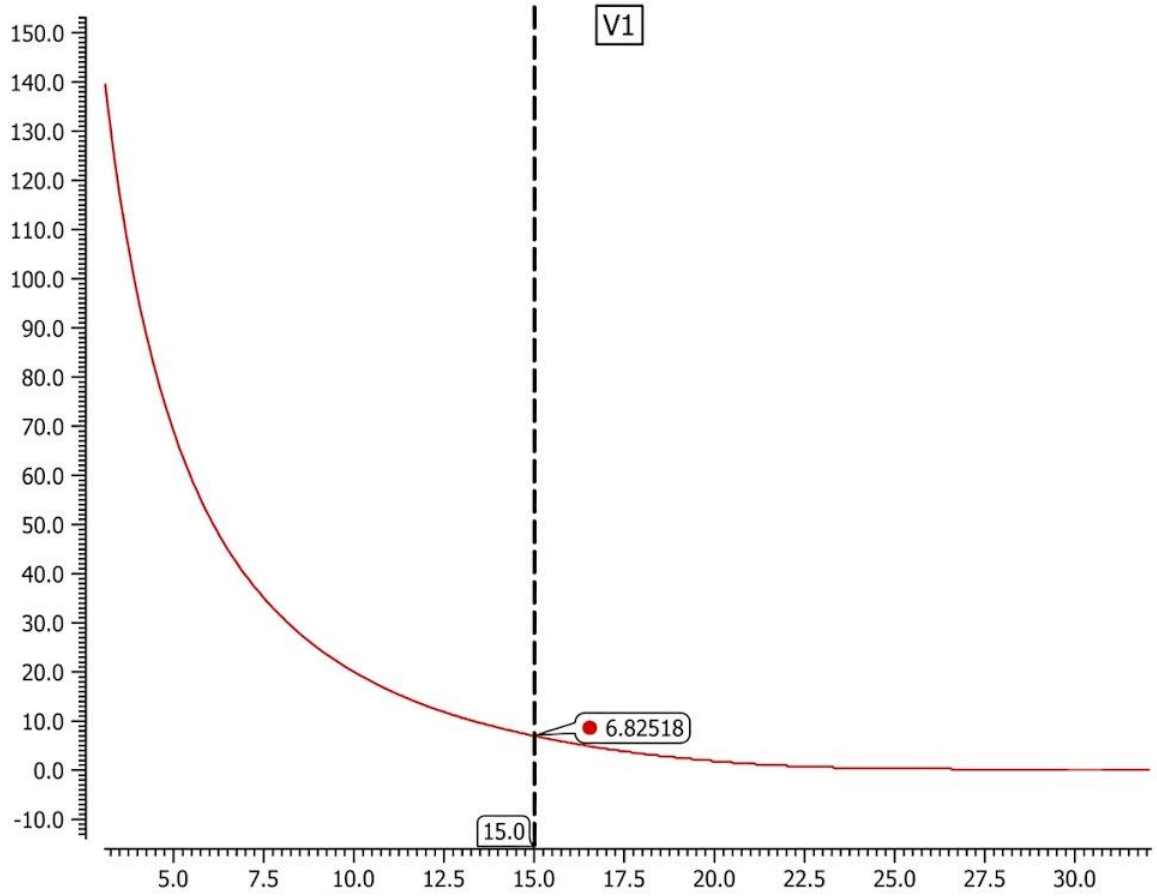
Id/W vs. gm/Id



ID = Drain current, W = Transistor width.

The **required transistor width** can therefore be calculated as:

W = ID/(ID/W)



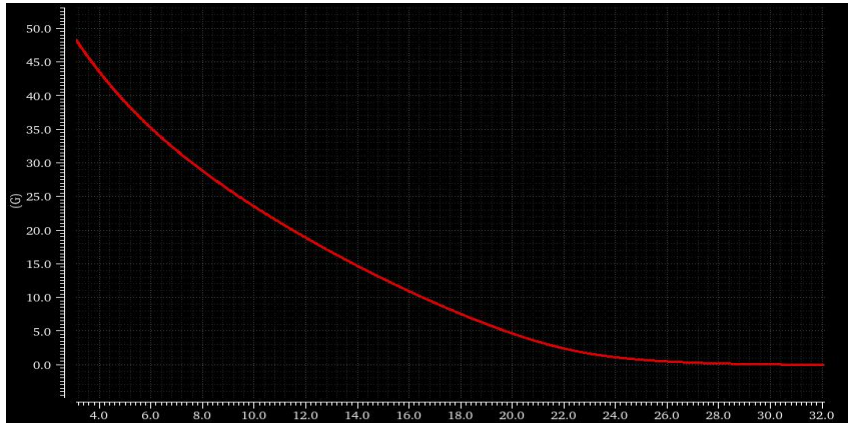
I_D vs. Transistor Width

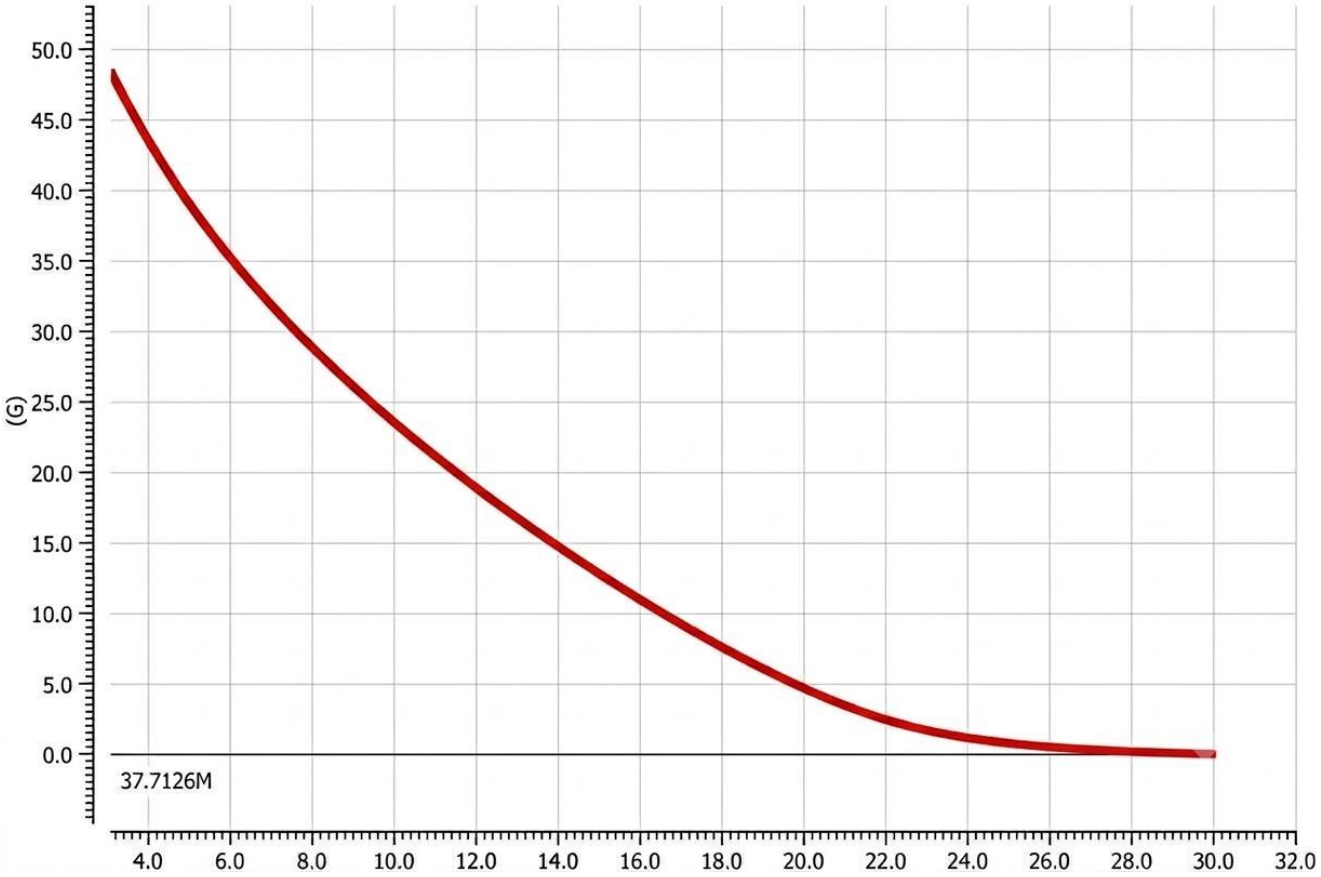
Fixed current density: $I_D/W = 6.82518\ \mu A/\mu m$

Drain Current, I_D	Current Density, I_D/W	Calculated Width, W
5 µA	6.82518 µA/µm	0.7326 µm
10 µA	6.82518 µA/µm	1.4652 µm
20 µA	6.82518 µA/µm	2.9306 µm
25 µA	6.82518 µA/µm	3.6628 µm
50 µA	6.82518 µA/µm	7.3251 µm
100 µA	6.82518 µA/µm	14.651 µm
200 µA	6.82518 µA/µm	29.301 µm
500 µA	6.82518 µA/µm	73.255 µm
1 mA	6.82518 µA/µm	146.408 µm

W = ID/(Id/w)

fT vs. gm/Id





The unity-gain frequency is approximately given by:

$fT = gm / (2\pi Cgg)$

Where, fT = Unity-gain frequency

gm = Transconductance

Cgg = Total gate capacitance $\pi = 3.14159$.

The total gate capacitance can be approximated as:

$Cgg \approx Cgs + Cgd$

Therefore: $fT \approx gm / [2\pi(Cgs + Cgd)]$

Oceanscript:

1. Simulator Setup & Workspace

```
simulator( 'spectre )
design("/home/m10/projects/Rakshitha/work_libs/m10/cds/simulation/nmos_char/spectre/schematic/netlist/netlist")
resultsDir( "/home/m10/projects/Rakshitha/work_libs/m10/cds/simulation/nmos_char/spectre/schematic" )
```

simulator: Initializes the Cadence Spectre simulation engine.
design: Specifies the exact path to the generated schematic netlist.
resultsDir: Defines the directory where all output simulation data will be saved.

2. Technology Models & Includes

```
modelFile( '("/opt/sclpdkv3/scl180/pdk/cdns/sclpdk_v3/HOTCODE/models/ts18scl/v2.0/hspice/ts18sl_scl.lib" "tt_18") )
definitionFile( "/home/m10/projects/Rakshitha/work_libs/m10/cds/Basics/nmos_char/save_op.scs" )
```

modelFile: Loads the SCL 180nm PDK device models, specifically selecting the "tt_18" (Typical-Typical) process corner.
definitionFile: Includes a custom setup file to force the simulator to save operating point (OP) parameters.

3. DC Sweep Analysis Configuration

```
analysis('dc ?dev "/V0" ?param "dc" ?start "0" ?stop "1" ?step "0.001" )
desVar( "vin" 0 )
envOption( 'analysisOrder list("dc") )
temp( 27 )
run()
```

analysis: Configures a DC sweep on the voltage source "/V0", sweeping it from 0V to 1V in 1mV steps.
desVar: Sets the default design variable "vin" to 0V.
envOption: Sets the simulator execution order, prioritizing the DC analysis.
run: Executes the simulation with all the defined parameters.

4. Core Parameter Extraction

```
Id = getData("M0:id" ?result "dc")
gm = getData("M0:gm" ?result "dc")
gm\/id = getData("M0:gmoverid" ?result "dc")
```

Extracts related parameters for transistor M0 from the DC results.

5. Derived Performance Metrics

```
FT = (getData("M0:gm" ?result "dc") / (2 * 3.14 * getData("M0:cgg" ?result "dc")))
Vov = (v("M0:vgs" ?result "dc") - v("M0:vth" ?result "dc"))
gm\/gds = (getData("M0:gm" ?result "dc") / getData("M0:gds" ?result "dc"))
Id\/w = (getData("M0:id" ?result "dc") / 1e-06)
```

6. Visualization & Plotting

```
ocnYvsYplot(?wavex gm\/id ?wavey FT)
```

ocnYvsYplot: Active command visualizing Transit Frequency (fT) on the Y-axis against Transconductance Efficiency (gm/Id) on the X-axis to analyze trade-offs

```
load("nmos_sim.ocn")
```

load: This command is typed into the Cadence CIW (Command Interpreter Window) to read, compile, and execute the entire OCEAN script file you have saved.